

office lighting is one of the top three demand-generation applications.

Approximately 66 per cent of survey participants (Fig. 2) also predict the increasing demand of LED for industrial areas including warehouses and manufacturing floors. In the hazardous work areas across industries, too, most of new lighting requirements are for LED lighting.

According to 60 per cent survey participants (Fig. 3), demand for LED lighting in India is still driven by metro cities due to affordability, awareness and socio-economic growth. However, 33 per cent survey participants also expect demand from Tier 2 cities, which will account for the major potential growth for streetlighting and industrial lighting. Only seven per cent of survey participants consider rural areas as the most demand-generating demography.

Technologies to shape the market

The advent of LEDs has transformed the way we see lighting. The very nature of LED lighting has opened up immense possibilities and has helped lower consumption levels, with the convenience of longer life and aesthetics that create an ambience that is welcoming and blends in effortlessly with the colour, mood, beauty and style of the place. Survey participants shared some insights about emerging technology trends that will shape the LED lighting market in India by creating more efficient, user-friendly products and enabling production techniques. Here is a collation of their views:

- Change in onboard technology through the use of IC based drivers to support touch based technology, which is the need of the hour for smart lighting systems. This, in turn, will reduce the number of compo-

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nents with respect to those used in traditional drivers. Use of lesser number of components can reduce cost while enhancing efficiency of the final product.

- Use of chip-scale packaging or Flip Chip packaging technology to enhance lumen output while increasing reliability of the final product. Use of chip-scale packaging eliminates the traditional submount to directly attach the LED die to the PCB, allowing for overall system cost reductions.

- Introduction of driverless low-voltage direct-current-operated products will enable energy saving by reducing AC-DC current conversion loss. This will also make the products compatible with solar photovoltaic systems and can run as LED-solar hybrid systems, which is quite effective for India.

- Shift in manufacturing techniques from through-hole to surface mount technology will enhance efficiency while reducing the operational cost. This, in turn, will help achieve breakeven point quickly despite relatively higher Capex investment.

What next

Like any other industry, sustainable growth of the LED lighting industry also depends on several factors. We

asked the survey participants to suggest possible challenges that could derail the growth of this industry in the country. Their collective opinions are as follows:

- Inability to make LED chips and micro-chips, resulting in import at high cost, also puts constraints in developing a wider variety and more colours, apart from enhancing performance and enabling innovations in LED lights

- Use of inefficient drivers resulting in

enhanced energy usage as well as product failure

- Acute shortage of experts in fields like chemical, electronics, optics, lighting and thermal management that are primarily required for manufacturing LED chips

- Mushrooming of low-quality production units, sub-standard quality and low cost of Chinese imports in India, which results in low consumer confidence

- Lack of awareness among consumers as well as institutional buyers involving government bodies about the efficiency of final products with respect to lux and watt, life expectancy and more, resulting in the use of products with higher wattage but lower efficiency

- Reduction of scope for domestic small/medium manufacturers with respect to government procurement tenders due to highly-stringent price caps, without mentioning enough quality parameters that can be achieved only by very large-scale production of inferior quality products or re-branding of low-cost import

Survey participants feel that successful resolution of the above-mentioned issues through appropriate industry initiatives and government interventions will help this industry to go ahead. **EFY**